PREMIER UNIVERSITY CHITTAGONG

Developed a 2D game using unity & C#

Course Code: CSE 456

Course Title: Computer Graphics & Image Processing Laboratory.

Submitted To

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Abstract

This lab report represents the design and development process for the game, created using Unity 2D and C#. The report begins by introducing the concept of game development. Then discusses the various components involved in game development, including game objects, scripts, scenes, and assets. The lab report also includes a step-by-step tutorial on how to create a simple game using Unity and C#. Finally, the report concludes with a discussion on the challenges and opportunities of game development with Unity and C#, as well as the potential for further exploration and experimentation in this field.

Contents

Chapter 1

Introduction

1.1 Introduction

This report describes the development process of a Santa Hill Climbing game created using Unity and C#. The game is a 2D plat former where the player controls Santa as he climbs a snowy hill, collecting presents and avoiding obstacles along the way. The game was developed as a project for a computer graphics and image processing Laboratory course in game design, with the aim of applying concepts to create an engaging game. Overall, the report demonstrates the application of programming and game design concepts in the development of a fully functional game.

Unity is a 2D/3D engine and framework that gives a system for designing game or app scenes for 2D and 3D. Unity allows us to interact with them via not only code, but also visual components, and export them to every major mobile platform and a whole lot more—for free. (There’s also a pro version that’s very nice, but it isn’t free. We can do an impressive amount with the free version.) Unity supports all major 3D applications and many audio formats, and even understands the Photoshop .psd format so we can just drop a .psd file into a Unity project. Unity allows us to import and assemble assets, write code to interact with your objects, create or import animations for use with an advanced animation system, and much more.

1.2 Background study

For this project we studied on various game design and development concepts from online resources. The design process involved considering the visual elements of the game, including the game environment, characters, and objects, as well as the gameplay mechanics, such as movement, obstacles, coin collecting system, pause-resume button system, coin count, and rewards etc. The development process involved using Unity and C# programming language to create the game mechanics and functionalities where includes such as object-oriented programming, collision detection, physics, and game mechanics were considered and implemented.

1.3 Necessary Technology

Here's a list of the necessary technology for the game:

- Unity game engine
- C# programming language
- Unity Editor Version
- Visual code editor

1.4 Related work

We find some project which related to our project. The development of the Santa Hill Climbing game was influenced by several related works in the game design and development field. Similar games such as Hill climbing, Endless runner, Super Mario, Sonic the Hedgehog, and Donkey Kong Country inspired the game's plat forming gameplay mechanics and level design.

Chapter 2

Methodology

In this game development process include the following these activities:

- **Requirements analysis**
  The game development starts with the analysis of the selected requirements. Here the requirement was to make a game 2d or 3d using unity and C# script and which assets need for our game.
- **Software Installation and asset collection**
  Next, install the necessary software, including the Unity game engine and C# programming language, Visual code editor, unity editor version latest or previous. And we collect all the assets which needs for our game.
- **Designing**
  Then we design our game using assets such as background, road, resume and pause button, vehicle and character, coin, jumping object etc.
- **Development**
  After complete the design process we implement all the functionalities using C# programming language. Here implement vehicle moving, background moving , camera moving, coin pick up , pause and resume button mechanism etc.
- **Testing**
  Testing is to check whether the flow of every function is correct or no by playing game.

Chapter 3

System Source code

3.1 For camera movement

```
using System.Collections;

using System.Collections.Generic;

using UnityEngine;


public class Camera1 : MonoBehaviour

{

    public Transform target;

    // private Vector3 offset;

    public float yoffset = 1f;
```

```
    public float FollowSpeed = 2f;

    // Start is called before the first frame update
    void Start()
    {
        // offset = transform.position - target.position;
    }

    // Update is called once per frame
    void Update()
    {
        // transform.position = target.position + offset;
        Vector3 newPos = new Vector3(target.position.x,target.position.y + yoffset,-10f);
        transform.position =Vector3.Slerp(transform.position,newPos,FollowSpeed*Time.deltaTime);
    }
}
```

3.2 For Vehicle

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.UI;

public class CarController : MonoBehaviour
{
    public static CarController instance;
    public float fuel = 1f;
    public float fuelconsumption = 0.1f;
    public Rigidbody2D carRigidbody;
```

```
public Rigidbody2D backTire;
public Rigidbody2D frontTire;
public float speed;
public float carTorque = 10;
private float movement;
public UnityEngine.UI.Image fuelimage;

public void Awake(){
    if(instance == null){
        instance =this;
    }
}
// Update is called once per frame
void Update()
{
    movement = Input.GetAxis("Horizontal");
    fuelimage.fillAmount = fuel;
}
private void FixedUpdate()
{
    if (fuel > 0)
    {
    backTire.AddTorque(-movement * speed * Time.fixedDeltaTime);
    frontTire.AddTorque(-movement * speed * Time.fixedDeltaTime);
    carRigidbody.AddTorque(movement * carTorque * Time.fixedDeltaTime);
    }
    fuel -= fuelconsumption * Mathf.Abs(movement) * Time.fixedDeltaTime;
}
```

```
}
```

3.3 For Fuel system

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

public class fuelFill : MonoBehaviour
{
    private void OnTriggerEnter2D(Collider2D collision)
    {
        CarController.instance.fuel = 1f;
        Destroy(gameObject);
    }
    void Start(){
    }
    void update(){
    }
}
```

3.4 For Pause and resume

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
public class PauseResume : MonoBehaviour
{
    public GameObject PauseScreen;
    public GameObject PauseButton;

    bool GamePaused;
```

```
// Start is called before the first frame update
  void Start()
  {
      GamePaused = false;
  }

  // Update is called once per frame
  void Update()
  {
      if (GamePaused)
          Time.timeScale = 0;
      else
          Time.timeScale = 1;
  }

  public void PauseGame()
  {
      GamePaused = true;
      PauseScreen.SetActive(true);
      PauseButton.SetActive(false);
  }

  public void ResumeGame()
  {
      GamePaused = false;
      PauseScreen.SetActive(false);
      PauseButton.SetActive(true);
  }
```

```
}
```

3.5 For Coin Score system

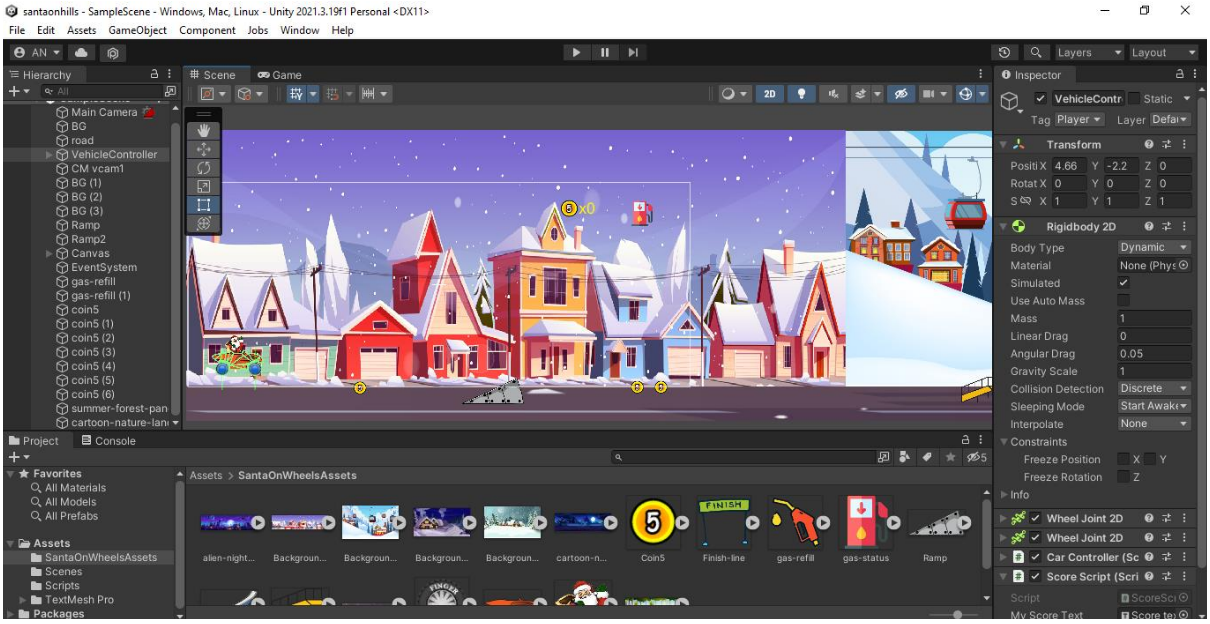
```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.UI;

public class ScoreScript : MonoBehaviour
{
    public Text MyScoreText;
    private int ScoreNum;
    // Start is called before the first frame update
    void Start()
    {
      ScoreNum = 0;
      MyScoreText.text = "X" + ScoreNum;
    }
    private void OnTriggerEnter2D(Collider2D Coin){
        if (Coin.tag == "mycoin")
        {
            ScoreNum = ScoreNum + 5;
            Destroy(Coin.gameObject) ;
            MyScoreText.text = "x" + ScoreNum;
        }
    }
}
```

Chapter 5

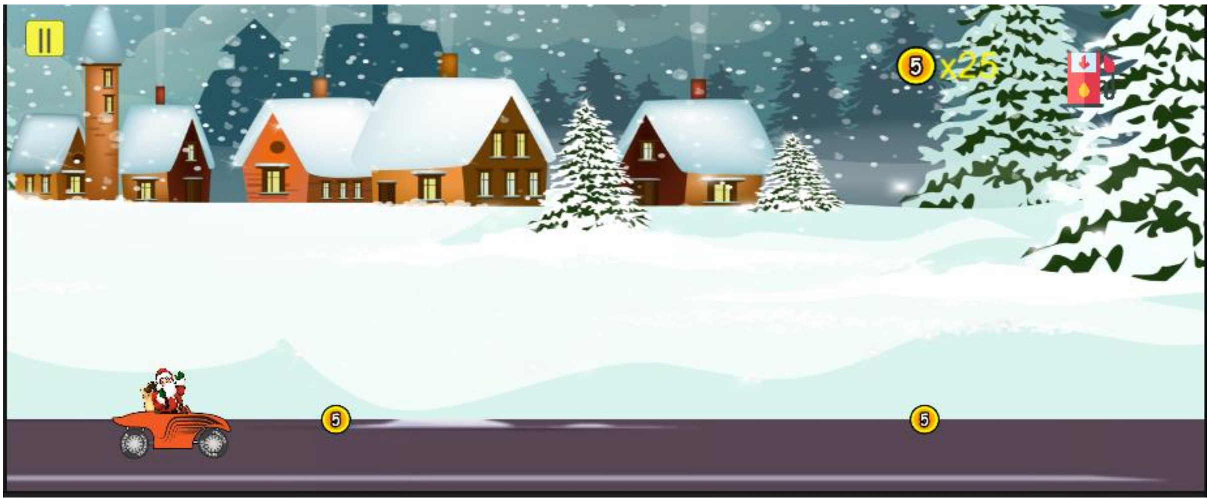
System Output

5.1 System Home



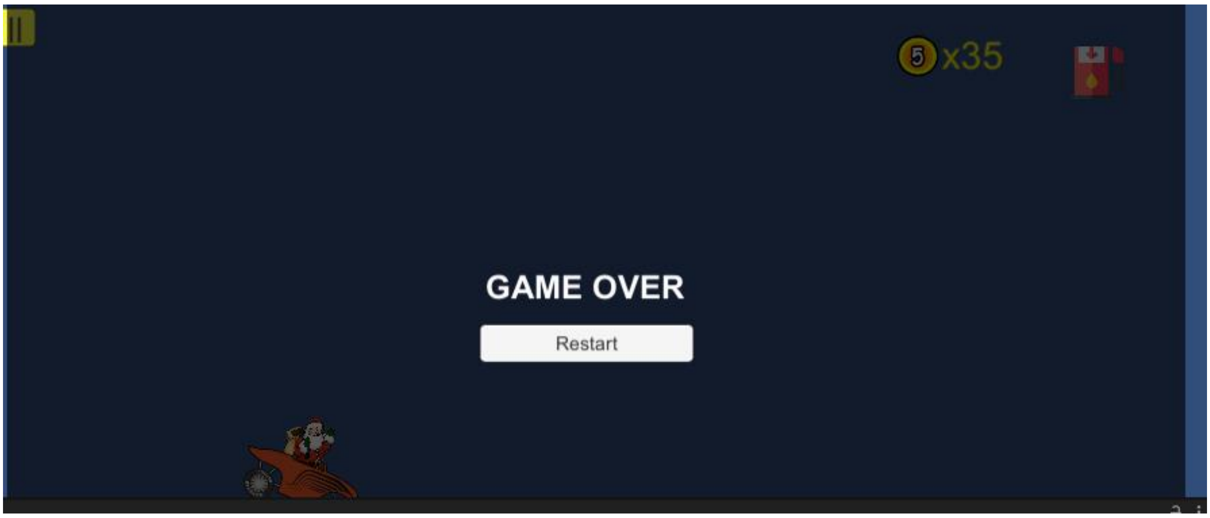
5.1 Game Play

5.1 Collecting coin & fuel



5.1 Pause and Resume

5.1 Game over



Chapter 6

Conclusion

6.1 Future Works

There are several potential areas for future work in the Santa Hill Climbing game:

- One area of improvement could be expanding the game's content with additional levels obstacles, and power-ups to keep players engaged over a longer period of time.
- Another area for improvement could be enhancing the game's visual and auditory elements such as adding more detailed textures and animations, and creating original soundtracks.
- Optimized for performance to ensure it runs smoothly on a wider range of devices, allowing more players to enjoy the game.

These future improvements could help increase the game's popularity and provide players with a more rewarding and enjoyable experience.

6.2 Conclusion

In conclusion, developing the Santa Hill Climbing game using Unity 2D and C# was an exciting and challenging project as a beginner that resulted in a fun and engaging game. The game's mechanics, level design, and winter wonderland atmosphere provide players with a unique and enjoyable experience. The development process involved careful planning, asset collection, and iterative development to ensure the game was polished and bug-free. The game's future work could include expanding the game's content, enhancing its visual and auditory elements, improving its scoring and leaderboard system, and optimizing it for performance on a wider range of devices. Overall, developing the Santa Hill Climbing game using Unity 2D and C# is an example of how technical expertise, creative design, and collaboration with C# and unity can come together to create a successful and enjoyable gaming experience.

Reference

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